Feature Selection in ML

A Deep Dive into Filter Methods, Correlation Analysis, and P-Value Hypothesis Testing

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Feature Selection Overview

What is Feature Selection?

Not all features in a dataset contribute positively to predictive modeling. Feature selection is the process of isolating the most relevant variables to build efficient machine learning models.

Quality over Quantity: Eliminating noisy predictors leads to higher model accuracy and generalization.

Reduces Dimensionality

Curse of dimensionality leads to sparse data distributions and high variance errors.

Efficient Computation

Smaller feature spaces dramatically reduce RAM usage and model training time.

Key Benefits of Selection

Faster Training

Lower dimensional data drastically reduces computational overhead, speeding up model training and real-time inference latency.

Reduced Overfitting

Eliminating redundant or noisy features prevents models from fitting random patterns, boosting out-of-sample accuracy.

Higher Explainability

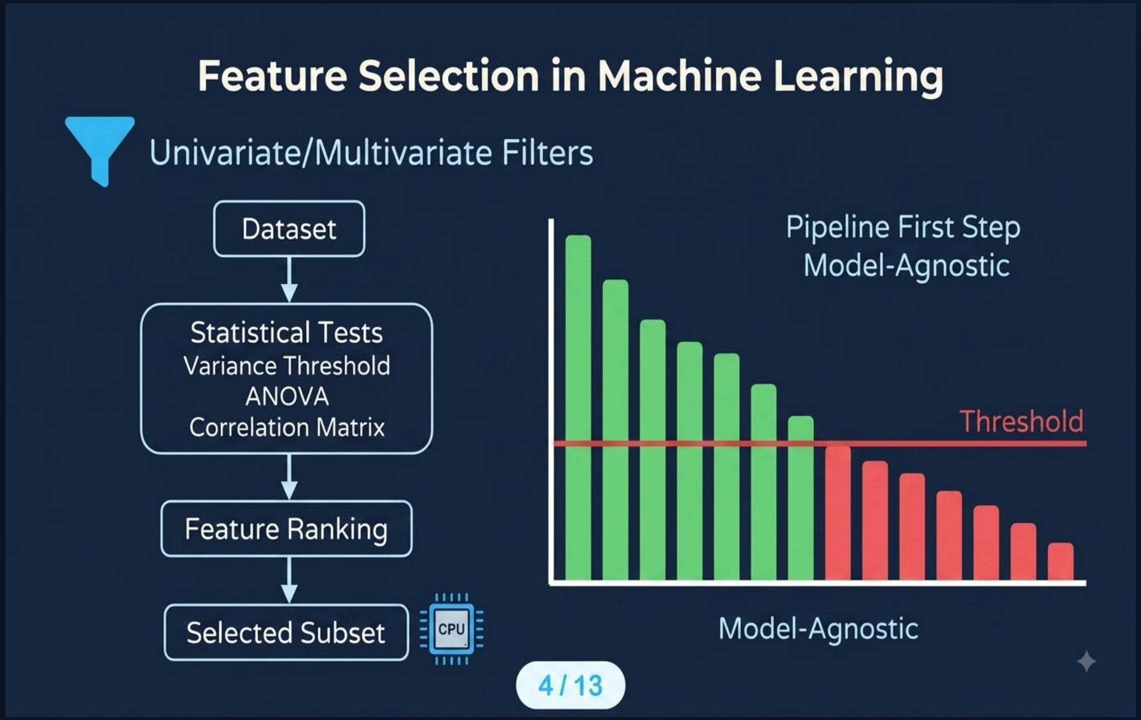
Models built with fewer, well-understood predictors are significantly easier to interpret, audit, and explain to executive stakeholders.

Understanding Filter Methods

Filter Method Workflow

Filter methods evaluate features prior to model training based purely on intrinsic statistical properties rather than predictive algorithms.

- **Model-Agnostic:** Independent of ML models, removing bias tied to specific estimators.
- **Pipeline First Step:** Highly computationally efficient pre-processing step for high-dimensional data.
- **Statistical Metrics:** Uses variance, correlation, ANOVA, and Mutual Information.



Univariate vs Multivariate Filter Methods

Univariate Filter Methods

Analyzes each variable individually against the target, ignoring interactions between features.

- Fast and scalable for massive datasets.
- Identifies zero variance and low correlation variables.
- Methods: Chi-Square, ANOVA F-test, Variance Threshold.

Multivariate Filter Methods

Evaluates dependencies and relationships across multiple features simultaneously.

- Detects feature redundancy and multi-collinearity.
- Prevents inclusion of duplicate information.
- Methods: Pearson Correlation Matrix, Mutual Information.

Removing Constant Features

Zero Variance Columns

Constant features contain identical values across every single sample in a dataset. Because they lack variation, they provide zero predictive capability.

- **Zero Signal:** Variance = 0 means no information is gained for classification or regression.
- **Automated Removal:** Easily dropped using Scikit-Learn's VarianceThreshold(threshold=0).

Scikit-Learn Implementation

```
from sklearn.feature_selection import VarianceThreshold

# Initialize selector
selector = VarianceThreshold(threshold=0.0)
X_high_variance = selector.fit_transform(X)
```

Correlation Thresholds

Linear Dependence & Redundancy

Pearson Correlation Coefficient (r) measures the linear association between continuous variables on a scale from -1 to +1.

- **Redundant Features:** When two variables exhibit $|r| > 0.85$, they supply nearly duplicate information.
- **Feature Dropping:** Safely drop one feature from each collinear pair to stabilize model weights and reduce collinearity.

$|r| > 0.85$

Standard Multi-Collinearity Cutoff

$r = 0.00$

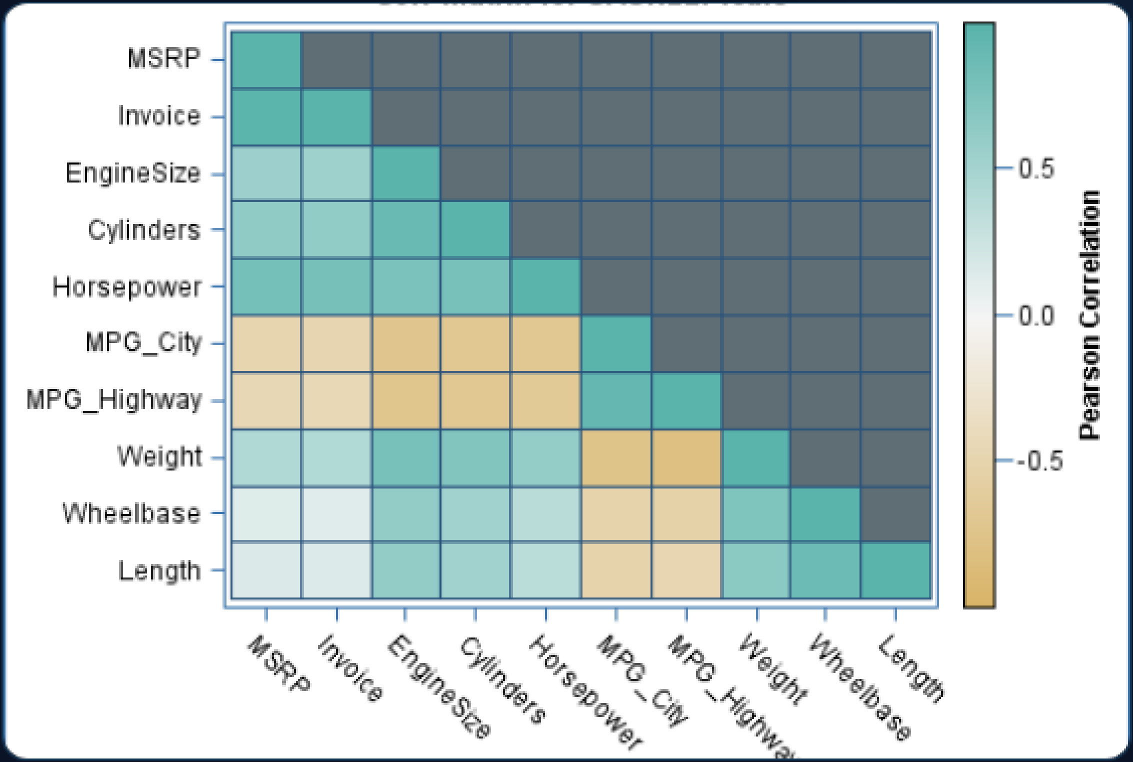
Zero Linear Association

Correlation Pair Analysis

Multivariate Correlation Heatmap

A correlation matrix visualizes pair-wise feature relationships, helping engineers quickly flag collinear predictors before training.

- **Heatmap Inspection:** Visual color gradients pinpoint high positive and negative correlations.
- **Deduplication:** Eliminates duplicate features while retaining maximum overall dataset variance.

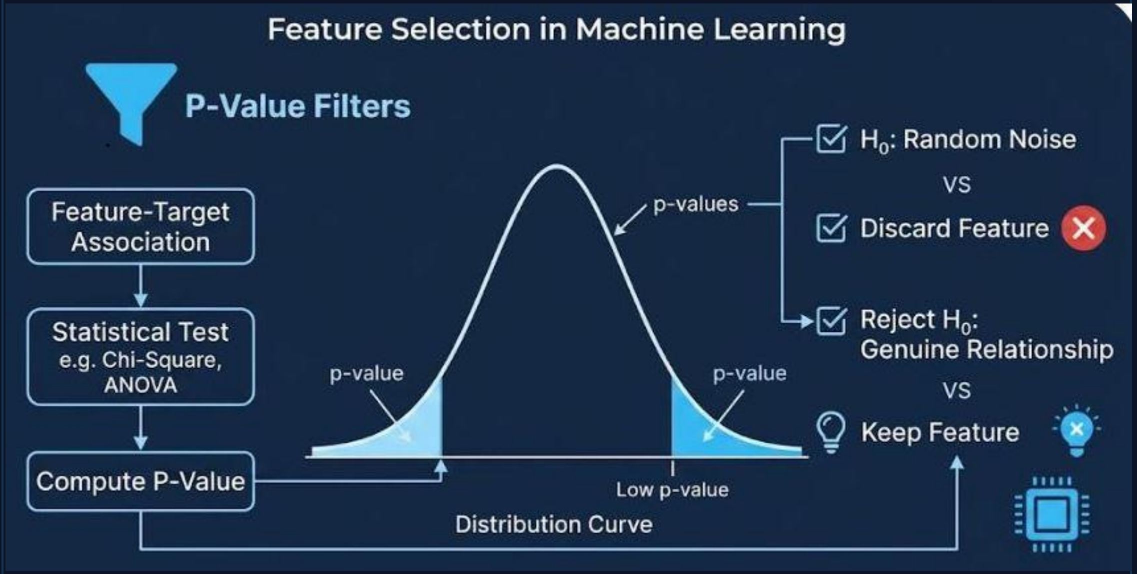


Role of the Null Hypothesis

Hypothesis Testing Framework

Statistical tests use null hypothesis testing (H_0) to establish whether a feature's association with the target is statistically meaningful or random noise.

- **Baseline Assumption (H_0):** Assumes no relationship exists between predictor variable and target outcome.
- **Mathematical Rigor:** Computes probability score (p-value) under H_0 distribution.
- **Feature Selection:** If H_0 cannot be rejected, the feature is deemed irrelevant and removed.



P-Value & Significance

$p \leq 0.05$

Statistically Significant

Reject the Null Hypothesis (H_0). The feature has a genuine relationship with the target and should be retained.

$p > 0.05$

Not Significant

Fail to reject Null Hypothesis (H_0). The observed pattern is likely random noise. Discard the feature.

Filter Method Comparison

Method	Type	Criterion / Statistical Test	Action Threshold
Variance Threshold	Univariate	Constant Column Variance	Drop columns where Variance = 0
Pearson Correlation	Multivariate	Linear Dependence ($\|r\|$)	Drop one feature if $\|r\| > 0.85$
ANOVA / Chi-Square	Univariate	P-Value Significance Test	Keep features with $p \leq 0.05$
Mutual Information	Multivariate	Information Gain Score	Select top-k highest scoring variables

Conclusion & Next Steps

Summary Checklist

- **Start with Filters:** Apply fast, model-agnostic filter methods early in your data pipeline.
- **Clean Noise:** Remove zero-variance constant features & highly collinear pairs ($|r| > 0.85$).
- **Validate Mathematically:** Use statistical p-value testing ($p \leq 0.05$) to confirm true feature significance.
- **Optimal Pipeline:** Achieve faster training, lower overfitting, and high model interpretability.

Next Steps in ML Pipeline

- **Wrapper Methods:** Experiment with Recursive Feature Elimination (RFE) for model-dependent tuning.
- **Embedded Methods:** Apply Lasso (L1) or Ridge (L2) regularization directly during model fitting.
- **Cross-Validation:** Always evaluate feature selection inside cross-validation folds to avoid data leakage.

References & Author Credits

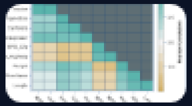
Documentation & Tutorials

- **Scikit-Learn Documentation:** Feature Selection Module (sklearn.feature_selection).
- **GeeksforGeeks:** Feature Selection Techniques in Machine Learning Pipelines.
- **SAS Statistical Graphics:** Correlation Matrix Analysis & Multicollinearity Heatmaps.
- **FU Berlin Statistics:** Hypothesis Testing, Normal Curves, and Critical P-Values.

Methodology & Attribution

- **Filter Method Theory:** Pearson Correlation Coefficient (r) & Variance Thresholding in High Dimensions.
- **Statistical Significance:** Hypothesis Testing Framework & Significance Cutoff ($\alpha = 0.05$).
- **Author Attribution:** Original presentation material created by **Venu Gopala Krishna Vanka**.

Image Sources



https://blogs.sas.com/content/sasdummy/files/2013/06/base_corrmatrix.png

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